COMBINED SOURCES SOUGHT NOTICE
AND NOTICE OF INTENT TO SOLE SOURCE

Announcement Type: Special Notice
Announcement Number: AMDTCNOI2601459
Subject: Highly Accelerated Stress Test (HAST) System for Reliability

Applicable NAICS: 334519 — Other Measuring and Controlling Device Manufacturing with a small business size standard of 600 employees.
Applicable PSC: 6640 – Laboratory Equipment and Supplies

Purpose
This is a combined sources sought notice and notice of intent to sole source, not a request for a quotation. NIST is seeking information from sources that may be capable of providing a solution that will achieve the essential requirements included in the "Description" section below.

If no alternate sources are identified, the Government intends to negotiate a firm-fixed price purchase order on a sole source basis with ESPEC North America, Inc., located at 4141 Central Pkwy, Hudsonville, Michigan, 49426-7828 USA. This action will be conducted under the authority of FAR RFO Part 12.102(a) and 41 USC 1901, allowing the Contracting Officer (CO) to solicit from one source. Delivery will be required within eight (8) weeks after contractor receipt of order.

Description
The Contractor shall deliver a quantity of one (1) Highly Accelerated Stress Test (HAST) System, inclusive of FOB Destination delivery and shall meet the minimum requirements identified below.

1. SYSTEM CONFIGURATION
   a) The system shall have a single cylindrical vessel.
   b) The system shall have multiple control modes including unsaturated, wet-saturated, and dry-wet bulbs to manage temperature and humidity.
   c) The system shall have touch-screen controller with temperature, humidity and count-down display and Ethernet interface.
   d) The system shall have minimum interior volume of 21 Liters.
   e) The system shall have weight not exceeding 200 kg.
   f) The system shall include water supply tank.
   g) The system shall have two stainless steel shelves.
   h) The system shall have status indicator light.
   i) The system shall have emergency stop switch.

2. VESSEL
   a) The vessel shall be able to accommodate at least one 200 mm wafer.
   b) The vessel shall have minimum interior diameter of 290mm.
   c) The vessel shall be able to be pressurized.
   d) The vessel shall be able to control both temperature and humidity.

e) The vessel and door shall be constructed with stainless steel.

3. OPERATION
   a) The system shall be able to operate at temperatures between 105°C to 162°C.
   b) The system shall be able to operate at humidity range between 75 to 100% RH.
   c) The system shall be able to operate at pressure between 0.020 – 0.392 MPa.
   d) The system shall have automatic water supply.
   e) The system shall be able to test samples with live load.
   f) The system shall have bias pins with 12 terminals.

4. CONTROL SYSTEM AND SOFTWARE
   a) The system shall be monitored and controlled remotely via ethernet connection.
   b) The system shall have the capability to view and edit recipes remotely.
   c) The system shall have the capability to display graphs on PC via ethernet connection.
   d) The system shall be able to email details on the alarms.

5. SAFETY
   a) The system shall have overcurrent protection.
   b) The system shall have overheat protector.
   c) The system shall have over pressure protector.
   d) The system shall have humidifier water level detection.
   e) The system shall have door lock safety mechanism to prevent opening of the door while the chamber is pressurized.
   f) The system shall have contract relay for external alarm.
   g) The system shall be able to shut down the product power in the event of an alarm.

6. CALIBRATION
   a) Vendor shall provide certificate of calibration.

Background

The National Institute of Standards and Technology (NIST), Division 735 at Engineering Laboratory, seeks information on commercial vendors that can provide a Highly Accelerated Stress Test (HAST) system to generate various defects for non-destructive defect detection metrology development. This is a CHIPS Metrology procurement supporting Project 3.08 for Grand Challenge 3, Enabling Metrology for Integrating Components in Advanced Packaging. The HAST test system significantly reduces the time required for humidity reliability testing, condensing weeks of conventional testing into hours or days. By simultaneously applying high temperature, high humidity, and high pressure, the HAST chamber accelerates moisture-induced failures. HAST will add new capability and productivity to produce defects in semiconductor packaging devices that will be used for defect characterization and failure analysis. The system, in support of NISTs mission, will be used for metrology development, including image acquisition, processing and optimization. This instrument will located in Building 220 in Gaithersburg, MD, and managed by the Engineering Laboratory.

The HAST test equipment significantly reduces the time required for humidity reliability testing condensing weeks of conventional testing into hours or days. By simultaneously applying high temperature, high humidity, and high pressure, the HAST chamber accelerates moisture-induced failures. The Non-Destructive Defect Detection Metrology for Semiconductor Advanced Packaging (NDDMAP) project team currently does not have HAST test equipment. This procurement is crucial for enabling NIST to establish non-destructive defect detection for semiconductor industry.

The CHIPS Metrology program develops and advances cutting edge metrology capabilities for members of the US semiconductor manufacturing ecosystem. This NIST conducted research program works with device manufacturers, tool vendors, materials suppliers, and other organizations to address critical metrology gaps to spur innovation within seven grand challenge areas. For more information on CHIPS Metrology, please visit https://www.nist.gov/chips/research-development-programs/metrology-program.

This is a CHIPS Metrology procurement supporting Project 3.08 for Grand Challenge 3, Enabling Metrology for Integrating Components in Advanced Packaging. The required HAST test system significantly reduces the time required for humidity reliability testing, condensing weeks of conventional testing into hours or days. By simultaneously applying high temperature, high humidity, and high pressure, the HAST chamber accelerates moisture-induced failures. HAST will add new capability and productivity to produce defects in semiconductor packaging devices that will be used for defect characterization and failure analysis. The system, in support of NISTs mission, will be used for metrology development, including image acquisition, processing and optimization. There are currently no other known sources capable of providing the required HAST System.

How to Respond to this Notice
Interested parties that believe they could satisfy the requirements listed above for NIST may clearly and unambiguously identify their capability to do so in writing by or before the response date for this notice. This is a combined sources sought notice and notice of intent to sole source is not a solicitation.

All responses to this notice of intent must be submitted via email to Jennifer Lohmeier via email at Jennifer.lohmeier@nist.gov so that they are received no later than **May 21, 2026 at 10:00 AM** Eastern Time.

Each response should include the following Business Information:

a. Contractor Name, Address,
b. Point of Contact Name, Phone Number, and Email address
c. Contractor UEI
d. Contractor Business Classification (i.e., small business, 8(a), woman owned, hubZone, veteran owned, etc.) as validated in System for Award Management (SAM). All offerors must have an active registration in www.SAM.gov.

e. Capability Statement describing what your company is capable of providing to meet the Government’s needs

Important Notes

The information received in response to this notice will be reviewed and considered so that the NIST may appropriately solicit for its requirements in the near future. This notice should not be construed as a commitment by the NIST to issue a solicitation or ultimately award a contract. This notice is not a request for a quotation. Responses will not be considered as proposals or quotations. No award will be made as a result of this notice. NIST is not responsible for any costs incurred by the respondents to this notice. NIST reserves the right to use information provided by respondents for any purpose deemed necessary and appropriate.